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PHYSICS
Paper 1
(Theory)
July/August 2026
2 1/2 hours

WAKISSHA JOINT MOCK EXAMINATIONS

Uganda Certificate of Education

PHYSICS

Paper 1

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

This paper has **two** *sections* **A** *and* **B.** *It has* **seven** *examination items.*

Section **A** *has* **three** *compulsory items.*

Section **B** *has* **two** *parts;* **I** *and* **II**. *Respond to* **one** *item from* **each** *part.*

Respond to **five** *items in all.*

Any additional item(s) responded to will not be scored.

All responses **must** *be written in the answer booklets/sheets provided.*

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Turn Over

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SECTION A

Respond to **all** *the items in this section.*

Item 1

A friend of your family, working abroad bought a plot of land next to your home. He has built a hotel next to your home. He intends to install security gadgets such as cameras and others. He wants to secure good glass for windows for nice appearance as well. He doesn't know how to select good glass and security cameras. He has contracted an expert to help in searching for good glass and security cameras with standard of 1.50 – 1.70 range of reflective index for glass and focal length of about 10·0 cm - 15.0 cm range when light of speed 3.0×10^8 ms^{-1} is used.

Hint: Speed of light in samples of glass given and power of the lenses present are as shown below.

glass	speed of light	Power of the Lens from glass material
A	1.97×10^8 ms^{-1}	8.3D
B	2.14×10^8 ms^{-1}	5.0D

In addition. when the glass was covered with gelatin layer of green, and viewed from inside the house. he observed features outside the premises appearing so different than when viewed from outside; which excited him.

Task

As student of physics help the friend to;

(a) (i) understand the meaning of refractive index of a substance and give the causes of refraction.

(ii) identify which type of glass should be taken on for use.

(iii) identify which type of lens should be selected for use in security cameras.

(b) Understand the meaning of dispersion and explain why objects were appearing different when seen from the inside and when viewed from the outside.

Item 2

Workers in a hospital are worried about the storage unit for a radioactive element used in treatment of cancer. They proposed to build a concrete bunker (barrier) in which to store it, but were not sure of it's dimensions. Upon consulting former employee of the Uganda Atomic energy council, he told them that radiation intensity, above 75 units is considered unsafe for humans.

The available data, indicating the relationship between thickness of concrete and the amount of the intensity that penetrates it is represented in the table below.

Intensity of radiations (units)	400	120	50	15	5.0
Thickness of concrete (cm)	0	10	20	30	40

Students who came to visit the hospital wondered why they needed to build a bunker and its exact dimensions. One of them even asked how these radioactive materials are used for detecting cancer in the body.

Task

Use your Knowledge of physics to help the student understand;

(a) the safe thickness of the concrete.

(b) why the bunker has to be constructed.

(c) how the radiations are used to treat cancer.

(d) how nuclear energy is produced in a nuclear plant.

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Item 3

A son who stays in South Africa's Cape Town invited his father for a visit. The son's home is near the ocean and on one evening the father saw the water level raising beyond the normal but later on went back. The old man got worried. While at his son's home, they were able to watch Uganda channels on television and the man wondered how signals could travel that far. One day he wanted to go to the mall for shopping, there was no body to escort him. But the son told him to use G.P.S to locate the mall and then back. He also wanted to know how it works.

Task

Using your knowledge of physics,

(a) Help the old man understand;
  (i) why water was raising beyond normal?
  (ii) the working of the G.P.S system.
(b) Explain to the old man some of the benefits of the raising water levels he observed.
(c) Describe how live broadcast is possible by television across different countries.

SECTION B

PART I

Respond to one item from this part

Item 4

A group of scouts on a camping trip noticed something peculiar about their flag at the lake shore. During hot days the flag flew towards land and at night, it flew towards sea even when there was no wind. They were puzzled by this observation. Later they met a local farmer who had recently built a dam on his farm. He was too unsure whether to fill the dam to its full capacity and why the walls of the dam were so thick at the bottom. Additionally the farmer was concerned about the quality of milk produced on his farm and suspected that his employees might be diluting it.

Hint: the dam is 8 metres deep and can support a maximum pressure of 70,000 Pa.

Density of water = 1000 kgm^{-3}

Task

As a student of physics;

(a) explain to the scouts why the flag behaved that way.
(b) help the farmer understand the purpose of thick walls at the bottom in the dam and advise him on filling the dam to capacity.
(c) suggest a way for the farmer to determine if the milk was diluted or if there were other factors affecting it.
(d) state the applications of the atmospheric pressure.

Item 5

In mountainous region, most people stay at the foot of the mountains while others stay at the top. People at the top experience water shortage and nose bleeding while those at the foot experience severe heat during dry spell. An NGO, a voluntary organization wants to help the people reduce on water shortage by installing a solar pump to provide constant water supply of 300 litres every minute but they do not know the height of the mountain. They have contacted a meteologist who gave them a barometer to help them in this task.

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Hint:
- Barometer gave that pressure difference between the top and bottom as 10 Kpa.
- Density of air is 1.25 kgm^{-3}.
- Acceleration due to gravity is 10 ms^{-2}.

Task

As a physics student, help the NGO and settlers to;

(a) explain why it is hotter at the bottom and cooler at the top.
(b) explain the causes of nose bleeding and water shortage at the top of the mountain.
(c) estimate the height of the mountain.
(d) determine the power of the pump required to raise water to the top of the mountain.

PART II

Respond to one item from this part.

Item 6

The district chairman is spear heading a project to start power generation from a certain water fall in the district to increase the supply, within the area. The project intends to produce 50,000 W at 500 V of a.c to be transmitted over a distance of 10 km across the district. The total resistance of the transmission cables that have been connected on the poles for transmission is 4 Ω per km. However the house hold appliances in the region requires a main supply of 240 V for proper use without any damage. The expert contracted also intends to sensitise the people on the recommended format and correct wiring system that will safeguard the devices connected on the grid.

Task

Using your knowledge of physics;

(a) help the chairman understand the advantage of generating A.C instead of D.C and guide him on the colour codes used in domestic wiring.
(b) explain to the chairman the working of the device that must be installed, through which the power must pass before it is connected for home use.
(c) determine the percentage power loss that occurs during the transmission.
(d) advise the chairman on ways how power loss during transmission can be reduced.

Item 7

A business man who operates a laundry business in one of the main trading centres in a district was charged Shs. 300,168 and is challenged with high electricity bill every month and feels cheated. He claims to operate two flat irons rated 1200 W each, for 2 hours per day, washing machine rated 2,200 W for 3 hours per day, and Three bulbs rated 100 W each for 10 hours per day. He thinks the rotating device in the washing machine over consumes power and therefore wishes to know how it operates.

Task

As a learner of physics;

(a) help the business man to know if he was being cheated or not
Hint: the unit cost of electricity is Shs.750
(b) Explain to the businessman to know how rotating device in the washing machine works.
(c) Advise on the possible means how these costs can be minimized.

END

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